

Student Name: Inayath Ali Mohammed

Student ID: 25736669

Written Reflection: A Comparative Life Cycle Cost Analysis of Sustainable and Conventional Buildings

(a) Actions Taken to Seek and Respond to Feedback:

In order to improve the analytical depth of my literature analysis, I actively sought input from peers and my academic supervisor throughout this research. According to early comments, my review mostly relied on broad assertions regarding the cost disparities between conventional and sustainable buildings, lacking adequate quantitative support and Sydney specific data. In response, I used empirical data from industry reports (GBCA, 2023) and peer reviewed studies (e.g., Zuo et al., 2017; Cabeza et al., 2014) to bolster claims that sustainable buildings achieve 10 – 20% lower overall life cycle costs over 30 – 50 years and 20 – 40% energy savings.

For instance, I created Table 1 (Literature Summary Table) that displays side by side cost components (original cost, energy cost, water cost, maintenance, total LCC) for sustainable vs. conventional structures in response to input regarding the need for more precise comparable measurements. This made it possible for me to transition from ambiguous assertions to evidence-based comparisons with literature-based percentage differences.

The significance of parameter sensitivity was further emphasized by supervisor input, especially with regard to discount rates (4 – 7%) and energy price escalation (2 – 5%). In response, I stated clearly the assumptions of my study (40 year lifespan, 5% discount rate, 3% energy escalation) and suggested sensitivity analysis as a primary goal. This improved the rigor and transparency of my study and immediately addressed the criticism that LCC outcomes can vary greatly in the absence of consistent assumptions.

(b) Difficulties Faced and Actions Taken to Overcome Them:

The absence of localized cost data for sustainable buildings in Sydney was a significant obstacle. The majority of previous research employed national or international averages, which made it challenging to directly apply to Sydney's particular building cost structure (6 – 10% yearly growth).

- **Action:** I highlighted this as a research gap in my paper and cited recent Australian studies, such as Infrastructure Australia's (2024) market capacity reports and ABS's (2024) construction price insights.
- **Result:** This strengthened the industrial relevance of my research for NSW and Sydney in particular while enabling me to accept the restriction.

Reconciling contradicting research on overall LCC savings was another challenge; some studies found reductions of 10 – 20%, while others showed lower amounts because of different discount rates.

- **Action:** To evaluate how changes in discount rates (4 – 7%) and increases in energy prices (2 – 5%) impact overall LCC outcomes, I created a sensitivity analysis methodology.
- **Result:** Instead of ignoring uncertainty, this allowed me to quantify it by turning a weakness into a structured analytical component. My third research question closely relates to the study's current focus on the sensitivity of LCC outcomes to financial conditions.

The requirement to gather and cross-validate secondary data from several sources (ABS, GBCA, journal papers) made time management difficult as well.

- **Action:** I developed a staged strategy that included the following steps:
 1. Gathering literature and identifying gaps
 2. Establishing assumptions and designing the LCC framework
 3. Conducting a comparative analysis
 4. Documenting and referencing.

- **Result:** I was able to incorporate peer and supervisor input iteratively without postponing submission thanks to this milestone-based methodology, which also guaranteed timely completion.

(c) Value Proposition for Potential Employers:

My capacity to apply life cycle costing (LCC), a fundamental economic evaluation method, to actual building design decisions has improved as a result of this project. I can now quantify trade-offs between long-term operational savings (20 – 40% in energy and water) and higher starting expenses (5 – 15% for sustainable buildings), and I can present these results as 10 – 20% reductions in total cost of ownership.

Employers in the fields of infrastructure policy, quantity surveying, construction economics, and sustainability consulting will respect my capacity to:

- When faced with uncertainty, apply conventional LCC techniques (discount rates, price escalation).
- Determine and fill up localized data gaps (analysis particular to Sydney).
- Convert sustainability aspects into measures of financial performance.

At this point in my career, I am well equipped to support evidence based choices that strike a balance between initial investment and lifecycle savings, which are essential for positions in infrastructure asset management, cost planning, and green building certification. I am certain that I can support groups that require rigorous analysis, useful sensitivity testing, and transparent sharing of long term financial results.